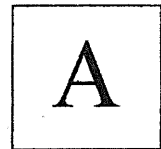


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B.Tech. Degree V Semester Special Supplementary Examination September 2022

ME 19-205-0504 THERMAL ENGINEERING
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Understand fuels, calorimetry, and basics of combustion of fuels.
 CO2: Determine the Air-Fuel ratio and analyse adiabatic flame temperature by applying thermodynamic laws to combustion.
 CO3: Study the air standard cycles used in various thermal equipments, its thermodynamic analysis, actual working of Internal Combustion Engines and its performance analysis.
 CO4: Understand various systems in Internal Combustion Engines components and alternate potential engines and their working.
 CO5: Study the different phases of combustion in Internal Combustion (IC) engines and also to understand the occurrence of combustion related phenomenon in IC engines.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome

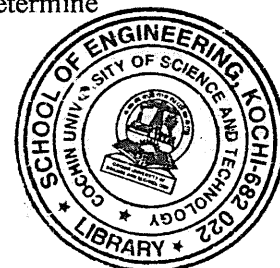
PART A(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Write short note on adiabatic flame temperature.		3	L2	2	1
(b) Write short notes on higher calorific value and lower calorific value.		3	L1	1	1,2
(c) Explain the difference between ideal and actual valve timing diagrams of a petrol engine.		3	L1	3	1,2,3
(d) What is meant by scavenging in I.C engines?		3	L2	3	1,2,3
(e) Write short notes on exhaust emissions of IC engines.		3	L1	4	1,2,3
(f) Explain Wankel engine with a neat sketch.		3	L2	4	1,2,3
(g) Write short notes on HUOCR.		3	L2	5	1,2,3
(h) Explain the various additives used in diesel engines.		3	L2	5	1,2,3

PART B

(4 × 12 = 48)

- II. The percentage composition by mass of a solid fuel is as follows: 12 L4 2 1,2
 C = 90%, H₂ = 3.5%, O₂ = 3%, N₂ = 1%, S = 1%, and the remainder being ash. Find:
- The mass of air required per kg of fuel for complete combustion.
 - The mass analysis of the dry products of combustion.
 - If 60% excess air is supplied in actual combustion, determine the mass of flue gases per kg of fuel.
 - Volumetric analysis of the products per kg of fuel.

OR

(P.T.O.)

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		Marks	BL	CO	PO
III.	A sample of gaseous fuel has the following composition by volume: $H_2 = 28\%$, $CO = 10\%$, $CH_4 = 2\%$, $CO_2 = 16\%$, $O_2 = 2\%$, $N_2 = 42\%$. If this fuel is burned with 50% excess air, determine: (i) the volume of air per cubic metre of gas at the same temperature and pressure (ii) the volumetric analysis of dry products (iii) the mass of products per kg of fuel (iv) the mass of dry products per kg of fuel (v) the mass of air supplied per kg of fuel.	12	L3	1	1,2,3
IV.	A 4-cylinder petrol engine has a bore of 60 mm and a stroke of 90 mm. Its rated speed is 2800 r.p.m. and is tested at this speed against brake which has a torque arm of 0.37 m. The net brake load is 160 N and the fuel consumption is 8.986 litres per hour. The specific gravity of petrol used is 0.74 and it has a lower calorific value of 44100 kJ per kg. A Morse test is carried out and the cylinders are cut out in the order 1,2,3,4 with corresponding brake loads of 110 N, 107 N, 104 N and 110 N respectively. Calculate for this speed: (i) Engine torque (ii) Brake mean effective pressure (iii) Brake thermal efficiency (iv) Mechanical efficiency (v) Indicated mean effective pressure. OR	12	L1	3	1,2,3
V.	The following data are known for a four-cylinder four-stroke petrol engine: cylinder dimensions: 11 cm bore and 13 cm stroke, engine speed: 2250 rpm, brake power: 50 kW, friction power: 15 kW, fuel consumption rate: 10.5 kg/hr, calorific value of fuel: 50,000 kJ/kg, air inhalation rate: 300 kg/hr and ambient condition: 15°C, 1.03 bar. Estimate: (i) brake mean effective pressure (ii) volumetric efficiency (iii) brake thermal efficiency (iv) mechanical efficiency.	12	L3	3	1,2,3
VI.	Explain the various ignition systems used in IC engines with the help of neat figure. OR	12	L3	4	1,2,3
VII.	Explain the following methods of governing I.C engines with a neat sketch: (i) Hit and miss method (ii) Quality governing (iii) Quantity governing	12	L2	4	1,2,3
VIII.	With a neat sketch explain the various combustion chambers used in S.I engines. OR	12	L2	5	1,2,3
IX.	With the help of a neat figure explain the stages of combustion in C.I engines.	12	L2	5	1,2,3

Bloom's Taxonomy Levels

L1 = 29.1%, L2 = 70.8%, L3 = 50%, L4 = 16.6%.
